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The Data Converter Family: A Typology of Architectures

Major ADC Architectures:

1. Flash: Parallel comparators for ultimate speed.

2. Pipelined: Staged conversion for high speed and good resolution.

3. Successive Approximation (SAR): Binary search algorithm for a balance of speed, resolution, and low power.

4. Sigma-Delta (ΣΔ): Over-sampling and noise shaping for the highest resolution at lower speeds.

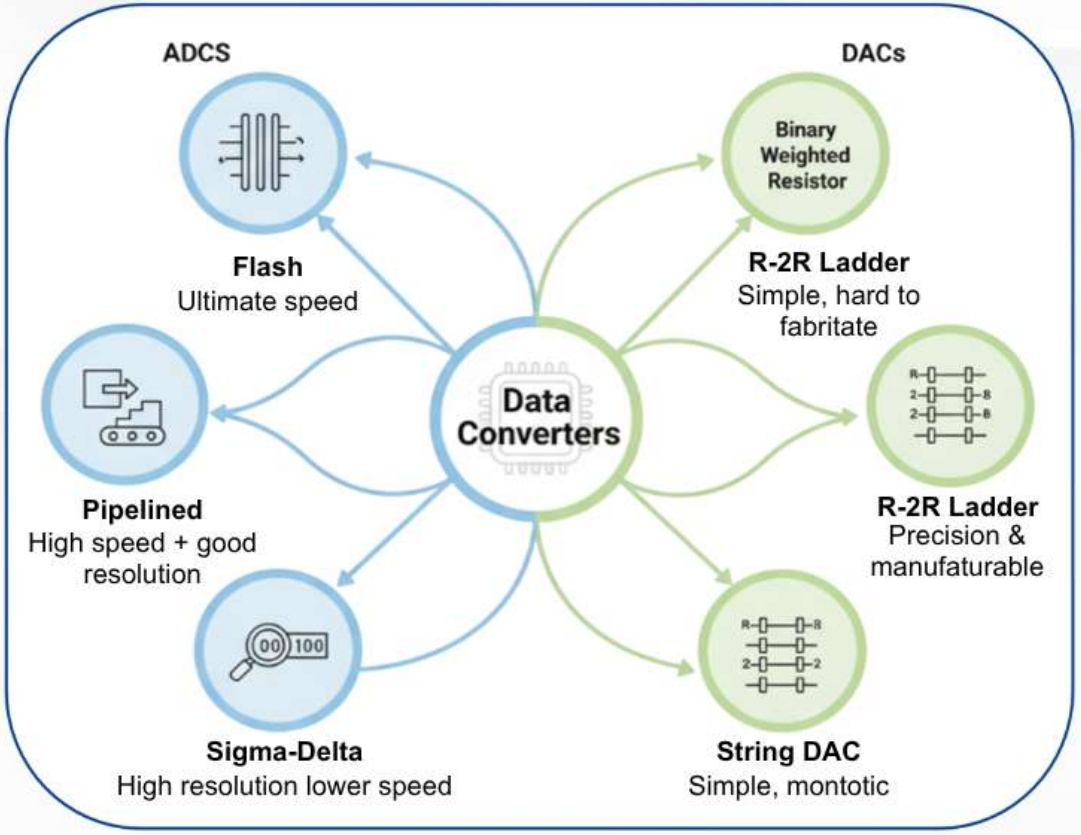
5. Dual-Slope (Integrating): High precision for DC measurements, but very slow.

Major DAC Architectures:

1. Binary Weighted Resistor: Conceptually simple but difficult to fabricate accurately.

2. R-2R Ladder: Uses only two resistor values, making it ideal for monolithic integration and high precision.

3. String DAC: Inherently monotonic, simple, used in digital potentiometers and as a sub-block in complex DACs.



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The Speed vs. Resolution Trade-off

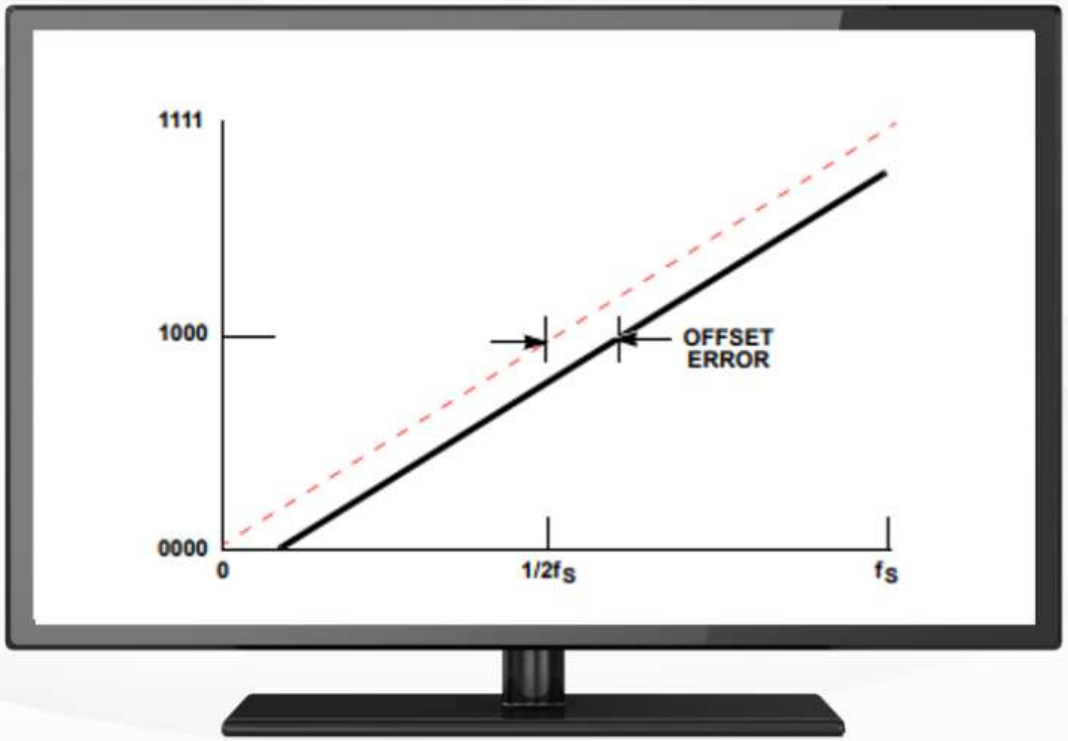
Architecture	Speed (Sample Rate)	Resolution (Bits)	Power/Cost	Key Feature/ Principle	Typical Application
Flash	Fastest (GSPS)	Low (≤12)	Very High	Parallel Comparators	Oscilloscopes, SDR
Pipelined	Very Fast (MSPS-GSPS)	Medium (12-16)	High	Staged Conversion	High-Speed DAQ, Comms
SAR	Medium (kSPS-MSPS)	Medium-High (12-20)	Low	Binary Search Algorithm	Industrial Control, BMS
Sigma-Delta (ΣΔ)	Slowest (SPS-kSPS)	Highest (16-32)	Very Low	Oversampling & Noise Shaping	Pro Audio, Precision Measurement
Dual-Slope	Very Slow (SPS)	High (≤22)	Low	Analog Integration	Digital Voltmeters

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Defining Performance: Key Static Metrics

Static metrics describe the converter's accuracy under DC or very low-frequency input conditions.



Resolution (N):

The number of bits in the digital code.

Defines the number of discrete quantization levels:2^N.

Determines the ideal step size, or Least Significant Bit (LSB).

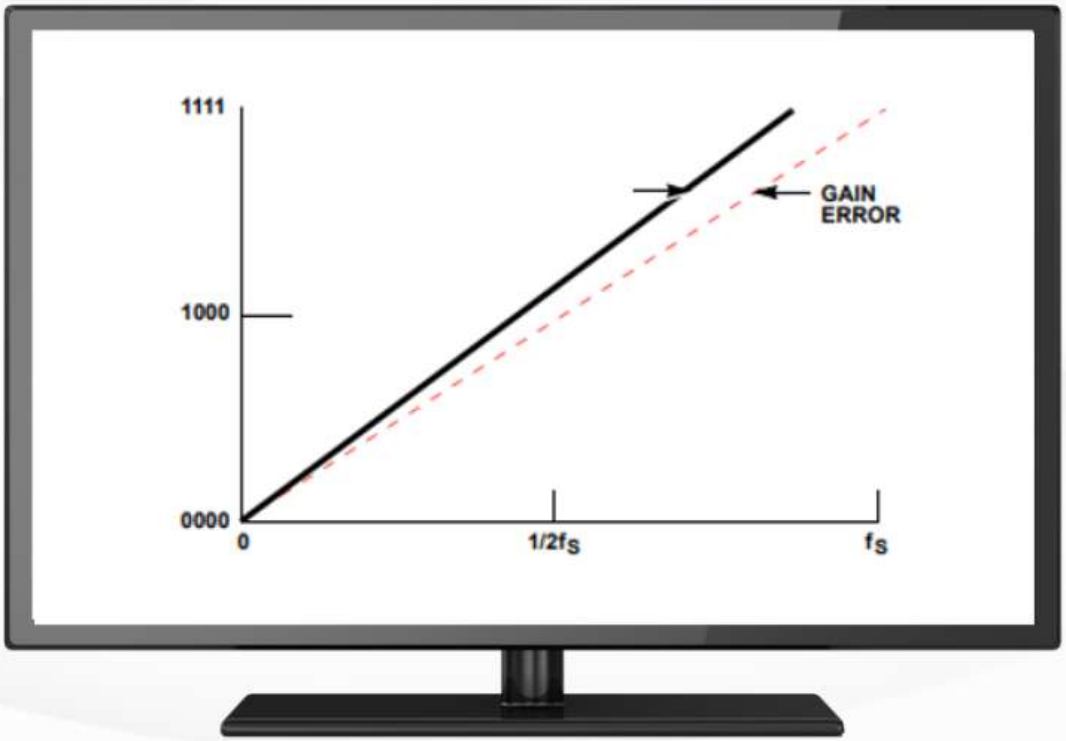
$$LSB = \frac{V_{REF}}{2^N}$$

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Defining Performance: Key Static Metrics

Static metrics describe the converter's accuracy under DC or very low-frequency input conditions.



Gain Error:

The deviation of the slope of the actual transfer function from the ideal slope.

Represents a scaling error that increases with input magnitude.

Total Unadjusted Error (TUE):

A comprehensive DC accuracy metric combining offset, gain, and linearity errors.

$$TUE = \sqrt{(OffsetError)^2 + (GainError)^2 + INL^2}$$

Static Errors: DNL and INL Explained

Integral Non-Linearity (INL):

- The maximum deviation of the actual transfer function from an ideal straight line, after offset and gain errors are nullified.
- Represents the cumulative effect of DNL errors: $INL[m] = \sum_{i=1}^{m-1} DNL[i]$
- **Significance:** INL is the primary measure of a converter's overall linearity and accuracy. It determines the distortion performance for AC signals.
- INL plots can reveal information about the ADC's internal architecture.

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Defining Performance: Key Dynamic Metrics

Signal-to-Noise Ratio (SNR):

- Ratio of the RMS signal power to the RMS noise power (excluding harmonics).
- Ideal N-bit SNR: $\boldsymbol{SNR\{ideal\} = 6.02N + 1.76\ dB}$

Total Harmonic Distortion (THD):

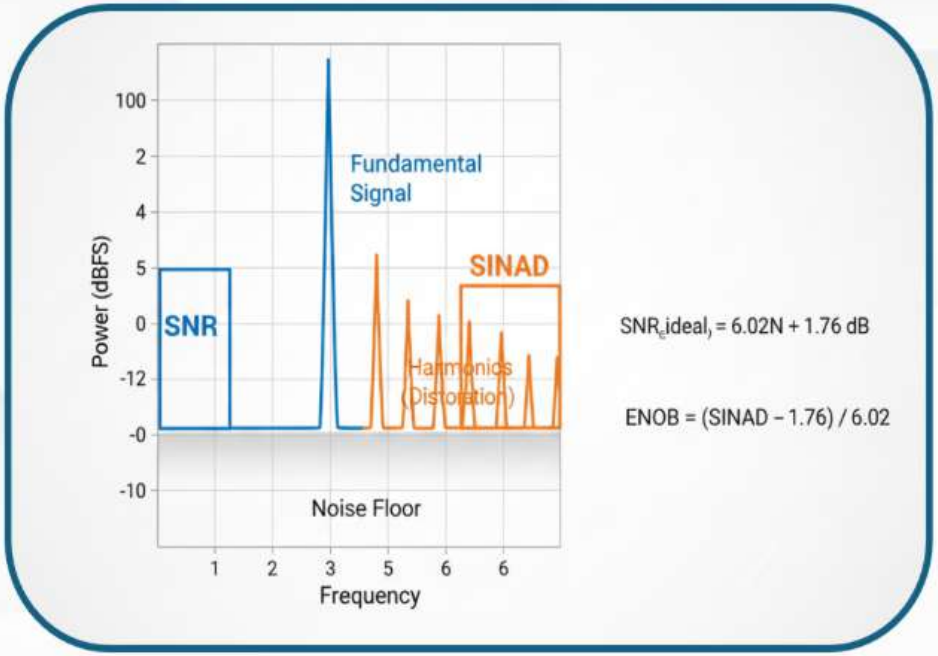
- Ratio of the RMS signal power to the RMS power of the harmonic distortion components. Caused by non-linearity (INL).

Signal-to-Noise and Distortion Ratio (SINAD):

- A comprehensive metric including both noise and harmonic distortion.
- $$SINAD = 10 \log_{10} \left(\frac{P_{Signal}}{P_{Noise} + P_{Distortion}} \right).$$

Signal-to-Noise and Distortion Ratio (SINAD):

- Effective Number of Bits (ENOB):
- $$ENOB = \frac{SINAD - 1.76}{6.02}.$$

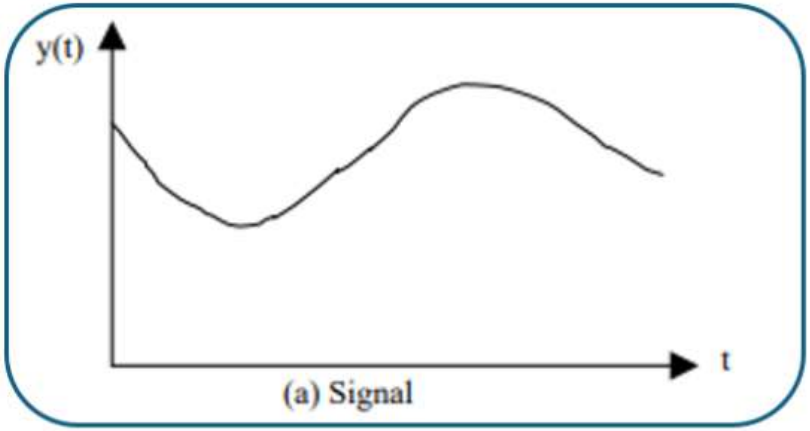


The Nyquist-Shannon Theorem & Aliasing

Click on the buttons to learn more.

The Foundation of Sampling:

- To accurately reconstruct a band-limited analog signal, the sampling frequency (f_s) must be at least twice the maximum frequency component (f_{max}) in the signal.
- is the Nyquist frequency.



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The Nyquist-Shannon Theorem & Aliasing

Click on the buttons to learn more.

What Happens When Nyquist is Violated?

- **Aliasing**: Frequencies in the input signal above the Nyquist frequency $f_s/2$ are not lost; they "fold back" or "alias" into the baseband **(0 to $f_s/2$).**
- **The Alias Frequency**: An input frequency **fin** greater than $\mathbf{f_s/2}$ will appear as a phantom frequency f_{alias} in the baseband, $where$ $f_{alias} = |k \cdot f_s - f_{in}|$, for an integer **k**.

Indistinguishable Error:

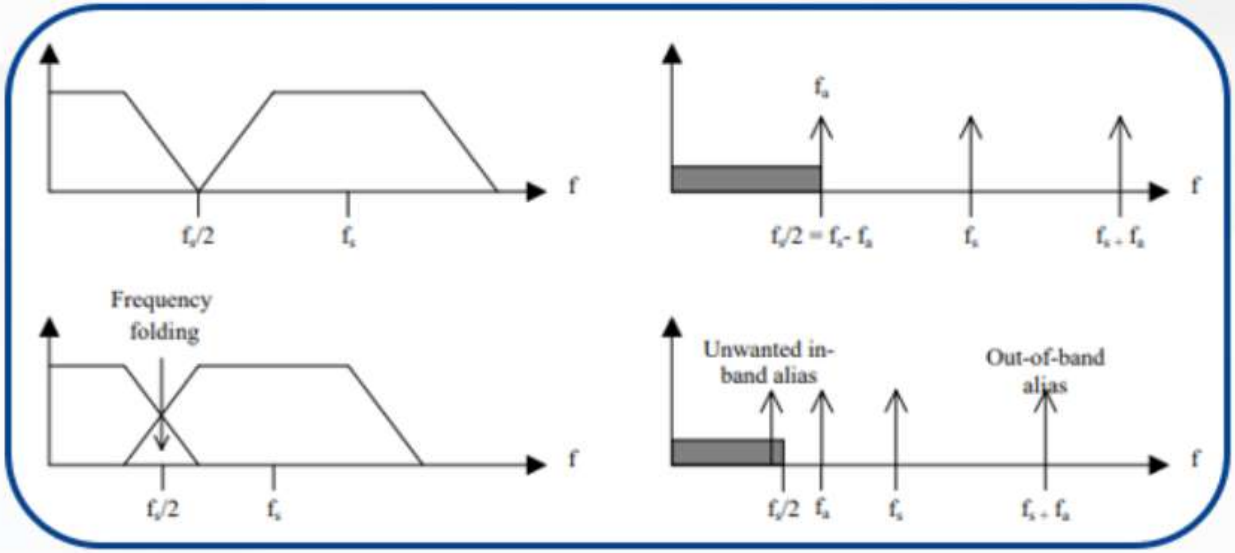
- Once aliased, this phantom frequency is mathematically indistinguishable from a real signal that was originally present at that frequency.

The Anti-Aliasing Filter (AAF)

Click the numbered buttons to learn more.

The Practical Solution to Aliasing:
Since we cannot control all frequencies present in a real-world signal, we must filter the input before it reaches the ADC.

Function: The Anti-Aliasing Filter (AAF) is a low-pass analog filter placed directly in front of the ADC's input.



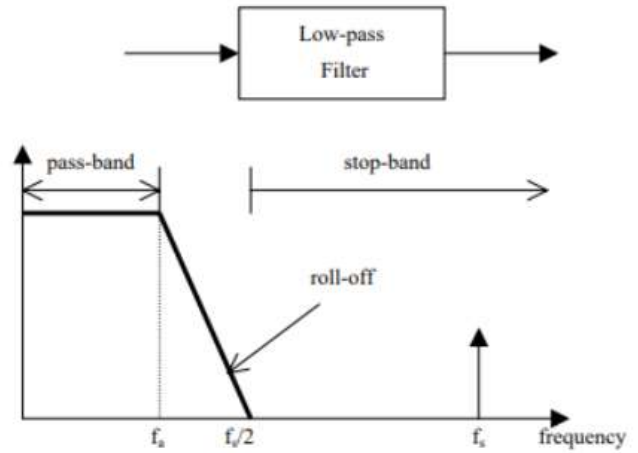
1 2

The Anti-Aliasing Filter (AAF)

Click the numbered buttons to learn more.

Design Trade-off:

- A higher sampling rate (oversampling) widens the gap between **fmax** and $f_s/2$.
- This "widens the transition band," allowing for a simpler, lower-order, and less expensive AAF.
- This is a key system-level design choice.



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The Workhorse: Successive Approximation (SAR) ADC

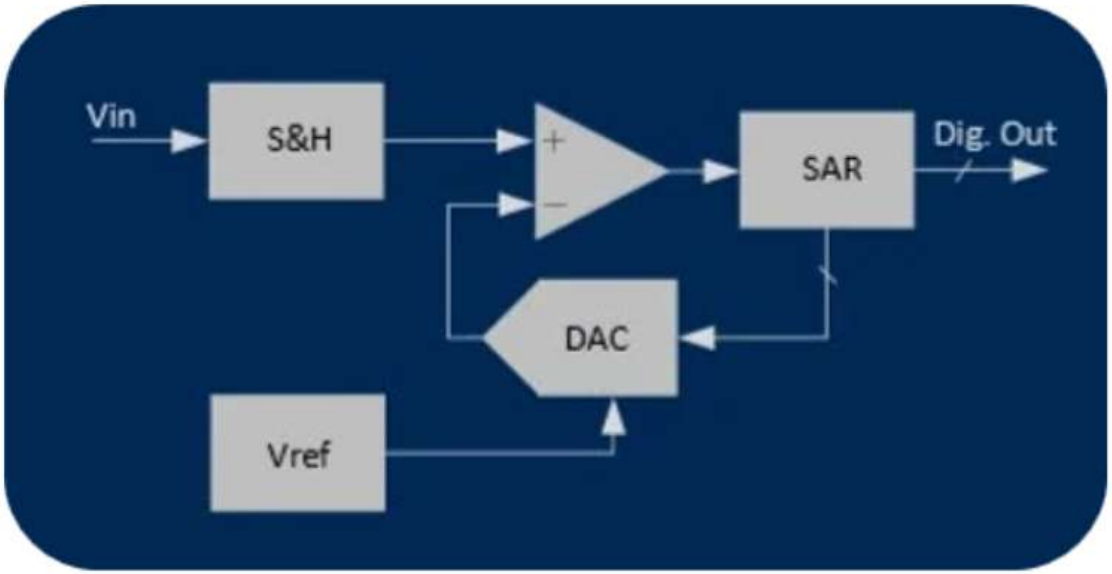
Core Components Block Diagram:

1. Sample-and-Hold (S/H) Amplifier: Acquires and holds the analog input voltage constant during conversion.

2. Analog Comparator: Compares the held input voltage to the DAC output.

3. Internal N-bit DAC: Generates an analog reference voltage based on the SAR's current digital guess.

4. Successive Approximation Register (SAR): The digital control logic that implements the binary search algorithm.



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SAR ADC: The Binary Search Algorithm

Operating Principle: The SAR ADC determines the digital code by performing a binary search over N clock cycles for an N-bit conversion.

Click the next arrow to view more categories.

1

Cycle 1 (MSB)

SAR sets MSB to 1, all others to 0 (Code: 100...0).

DAC outputs $V_{REF}/2$.

Comparator: If $V_{IN} > V_{REF}/2$, **MSB** remains 1. Else, **MSB** is cleared to **0**.

2

Cycle 2 (MSB-1)

SAR sets **bit (N-2)** to 1 **(Code: $b_{N-1}10$...).**

DAC outputs a new voltage based on the updated code.

Comparator: decides if bit **(N-2)** should be kept or cleared.

3

Cycles 3 to N (LSB)

Process repeats, one bit per clock cycle, moving from MSB down to LSB.



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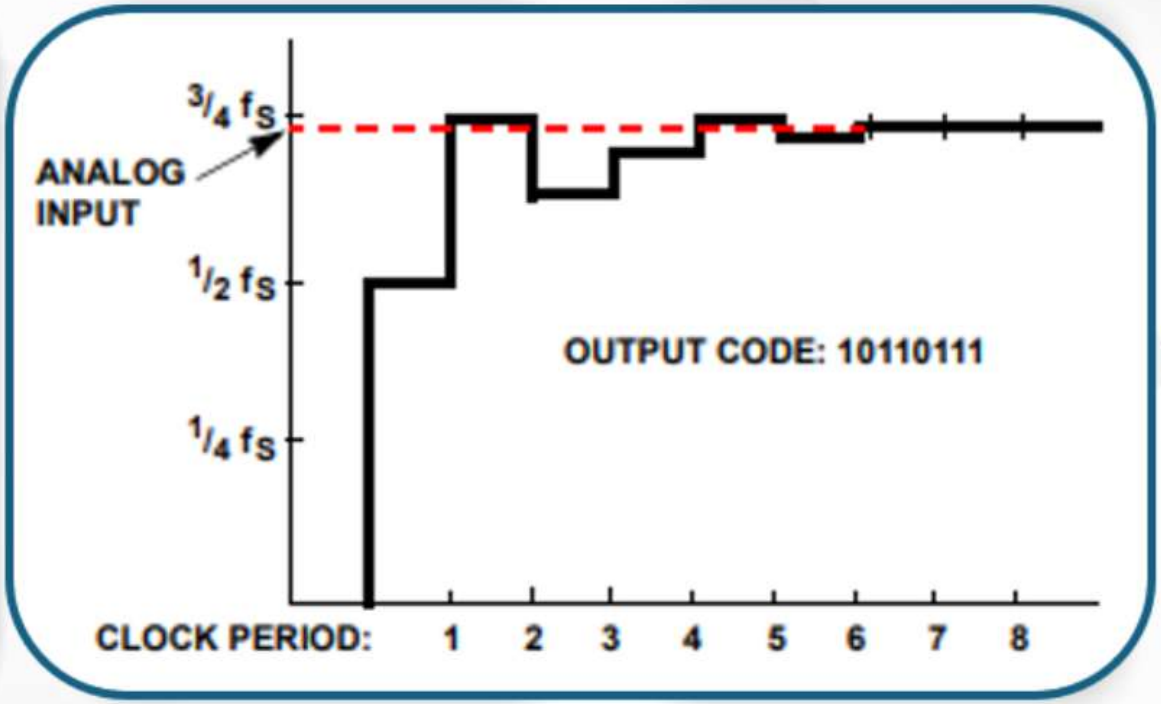
SAR ADC: The Binary Search Algorithm

Operating Principle: The SAR ADC determines the digital code by performing a binary search over N clock cycles for an N-bit conversion.

4

End of Conversion (EOC):

After N cycles, the SAR holds the final N-bit digital result.



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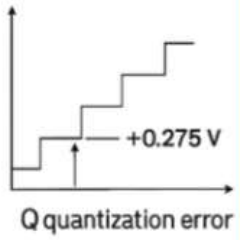
SAR ADC: Calculation Example (4-bit)

Conversion Steps:

1. **Cycle 1 (MSB, b3):**
   - Test Code: 1000 → $V_{DAC} = 5V \cdot (1/2) = 2.5\ V$
   - Comparison: 3.4V>2.5V
   - Result: Keep b3=1. Current Code: 1xxx
2. **Cycle 2 (b2):**
   - Test Code: 1100 → $V_{DAC} = 5V \cdot (1/2 + 1/4) = 3.75\ V$
   - Comparison: 3.4V<3.75V
   - Result: Clear b2=0. Current Code: 10xx
3. **Cycle 3 (b1):**
   - Test Code: 1010 → $V_{DAC} = 5V \cdot (1/2 + 1/8) = 3.125\ V$
   - Comparison: 3.4V>3.125V
   - Result: Keep b1=1. Current Code: 101x
4. **Cycle 4 (LSB, b0):**
   - Test Code: 1011 → $V_{DAC} = 5V \cdot (1/2 + 1/8 + 1/16) = 3.4375\ V$
   - Comparison: 3.4V<3.4375V
   - Result: Clear b0=0. Current Code: 1010

Resolution (N) = 4 bits
Reference Voltage (V_{REF}) = 5 V
Input Voltage (V_{IN}) = 3.4 V
LSB Size = 0.3125 V

Cycle	Bit Under Test	SAR Test Code	VDAC Calculation	Comparison
1	b_3	1000	2.5 V /	$V_{IN} > V_{DAC}$
2	b_2	1100	3.75 V	$V_{IN} < V_{DAC}$
3	b_1	1010	3.125 V	$V_{IN} < V_{DAC}$
4	b_0	1011	3.125 V	Keep Clear



Final Result: Digital Output = 1010b (Decimal 10).

Analog Value = $10 \times 0.3125V = 3.125V$.

Quantization Error = $3.4V - 3.125V = +0.275V$.

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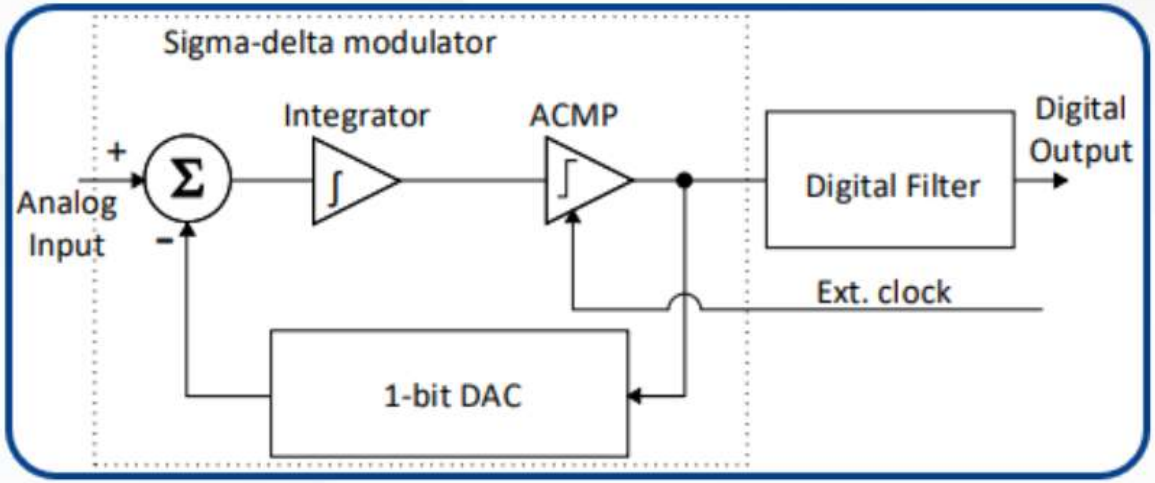
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The High-Resolution Specialist: Sigma-Delta (ΣΔ) ADC

Click the numbered buttons to learn more.

Key Characteristics:

- **Resolution**: Highest available (16 to 32 bits).
- **Speed**: Low output data rates (SPS to hundreds of kSPS).
- **Linearity**: Inherently monotonic due to the 1-bit conversion stage.
- **Filtering**: Inherent anti-aliasing due to oversampling.

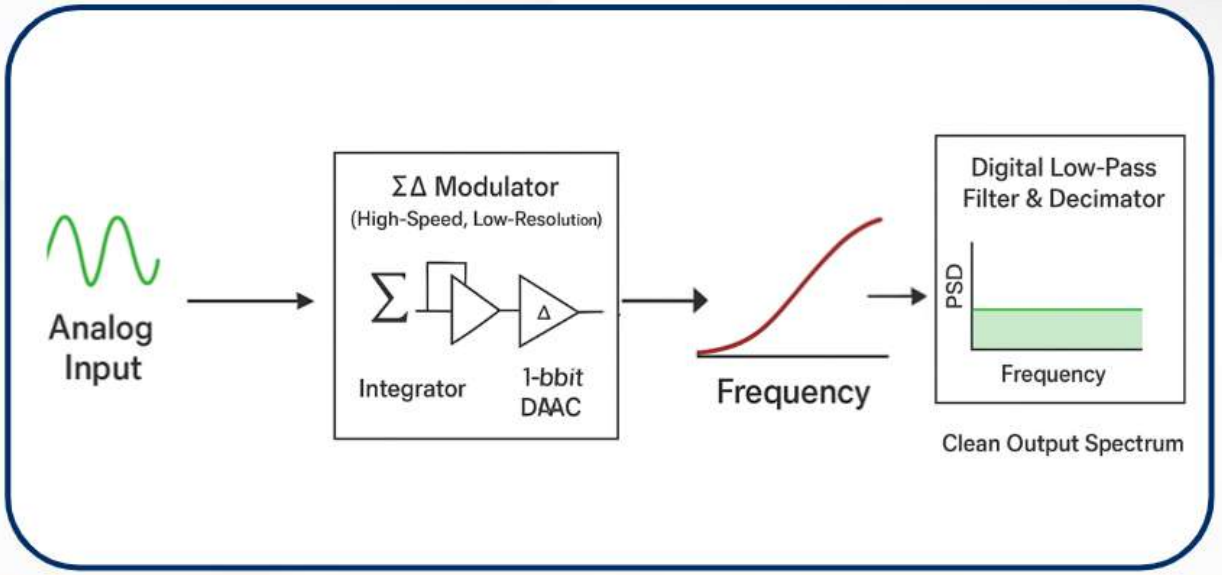


1 2

The High-Resolution Specialist: Sigma-Delta (ΣΔ) ADC

Core Principles (Different from Nyquist Converters):

- **Oversampling**: The analog input is sampled at a very high frequency f_{MOD}, many times the final output data rate f_{DATA}.
- **Noise Shaping**: A feedback loop with an integrator "shapes" the quantization noise, pushing it to higher frequencies, away from the signal band of interest.
- **Digital Filtering & Decimation:** A complex digital filter removes the high-frequency noise and then downsamples (decimates) the data to the final output rate, which increases the effective resolution.



ΣΔ ADC: Oversampling and Noise Shaping

Oversampling:

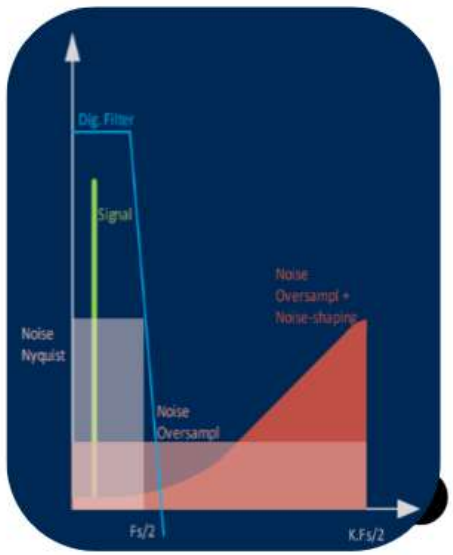
Sampling at a rate $k \cdot f_s$, where k is the Oversampling Ratio (OSR).

The total quantization noise power ($q^2/12$) remains constant, but it is now spread over a much wider bandwidth (0 to $k \cdot f_s/2$).

The noise within the original signal bandwidth (0 to fs/2) is reduced by a factor of k.

SNR improvement from oversampling alone: $SNR \propto 10\log_{10}(k)$
Doubling the OSR improves SNR by 3 dB (0.5 bits).

1



Click on the graph image to enlarge.

2

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ΣΔ ADC: Oversampling and Noise Shaping

Oversampling:

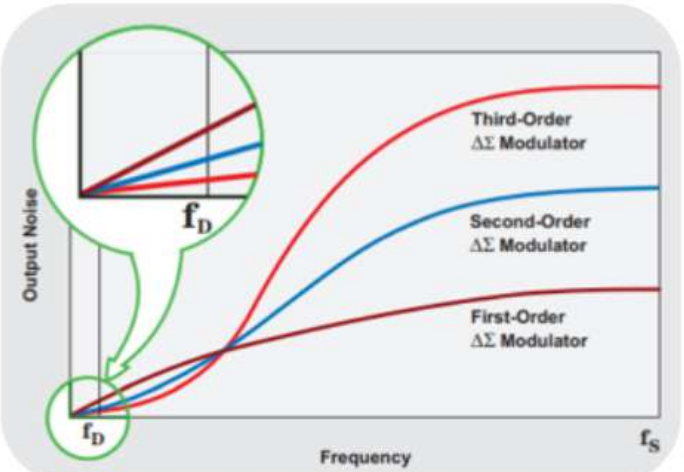
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The noise within the original signal bandwidth (0 to fs/2) is reduced by a factor of k.

SNR improvement from oversampling alone: $SNR \propto 10\log_{10}(k)$ Doubling the OSR improves SNR by 3 dB (0.5 bits).

1



Click on the graph image to enlarge.

2

Noise Shaping:

The modulator's feedback loop acts as a high-pass filter for quantization noise and a low-pass filter for the signal.

The noise is "shaped," pushing most of its energy to high frequencies.

For a first-order modulator, the noise power in-band is drastically reduced.

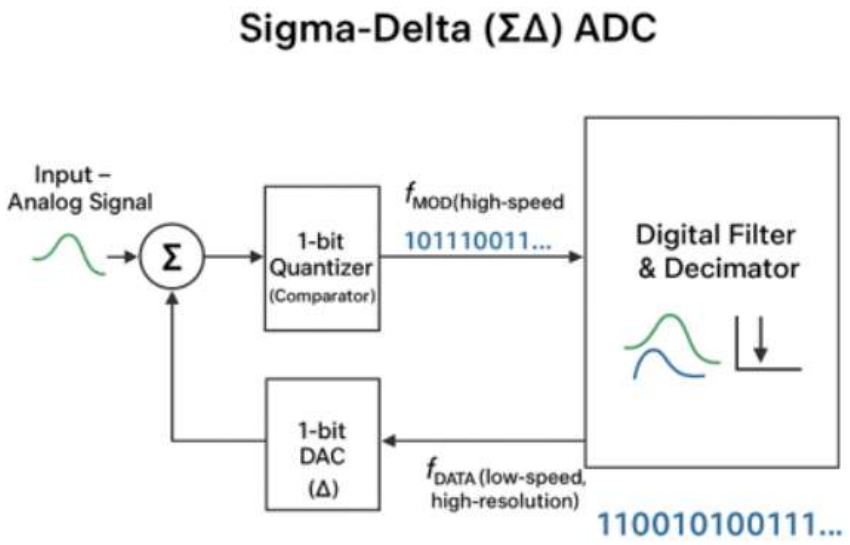
SNR improvement from a first-order modulator: $SNR \propto 30\log_{10}(k)$ .Doubling the OSR improves SNR by 9 dB (1.5 bits).

Higher-order modulators provide even more aggressive noise shaping.

ΣΔ ADC: Modulator and Digital Filter

The Digital Filter and Decimator:

- This is a computationally intensive digital signal processing block.
- **Function 1 (Filtering):** Implements a steep low-pass filter to remove the high-frequency quantization noise shaped by the modulator. Common types are Sinc filters.
- **Function 2 (Decimation):** Reduces the sampling rate from the high modulator frequency f_{MOD} down to the final output data rate f_{MOD},effectively trading speed for resolution.
- The final output is a high-resolution, Nyquist-rate data word.



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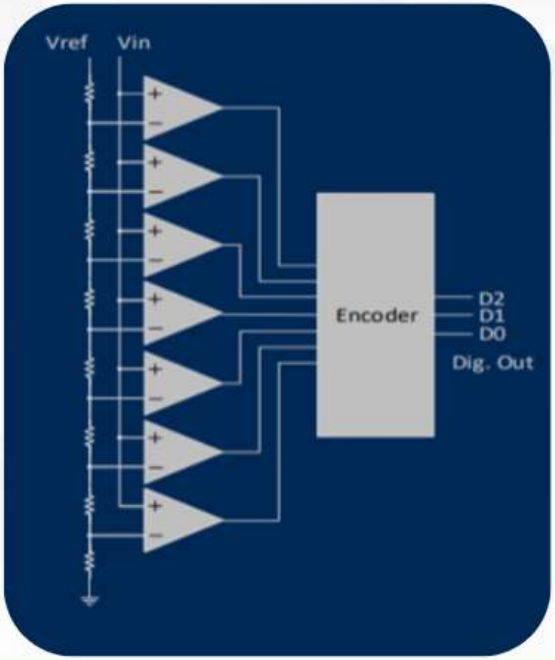
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The Speed Demon: Flash ADC

Click the numbered buttons to learn more.

Operating Principle: Direct conversion in a single clock cycle.

- A resistor string divides the reference voltage V_{REF} into $2^N - 1$ unique reference levels.
- An array of $2^N - 1$ comparators simultaneously compares the input voltage (V_{IN}) against each reference level.
- The output of the comparators forms a "thermometer code" (e.g., 00011111).
- A priority encoder converts the thermometer code into the final N-bit binary output.



1 2